Exam 1

ECE 2201: Circuit Analysis I

Summer 2025

June 24, 2025

1. This exam is closed book, closed notes and a crib sheet has been provided to you with the exam. Print your name below.
2. Show all work on these pages. Show all work necessary to complete the problem. A solution without the appropriate work shown will receive no credit. A solution which is not given in a reasonable order will lose credit.
3. Show all units in solutions, intermediate results, and figures. Units in the exam will be included between square brackets.
4. If the grader has difficulty following your work because it is messy or disorganized, you will lose credit.
5. Do not use red ink. Do not use red pencil.
6. You will have 75 minutes to work on this exam.

Name: Solution ID: __________

1.________ /25
2.________ /25
3.________ /25
4.________ /25

Problem 1

A Device shown in Figure A has a current, $i_d(t)$, and a voltage, $v_d(t)$, as shown in Figure B.

a. Write the piece-wise functions for the current, $i_d(t)$, and a voltage, $v_d(t)$.
b. How much charge has moved through the device for $t > 0$.
c. Make a graph of power vs. time for $t > 0$.
d. How much energy was supplied by the device for time $t > 0$.

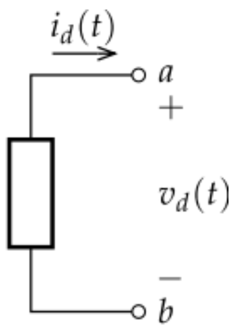


Figure A

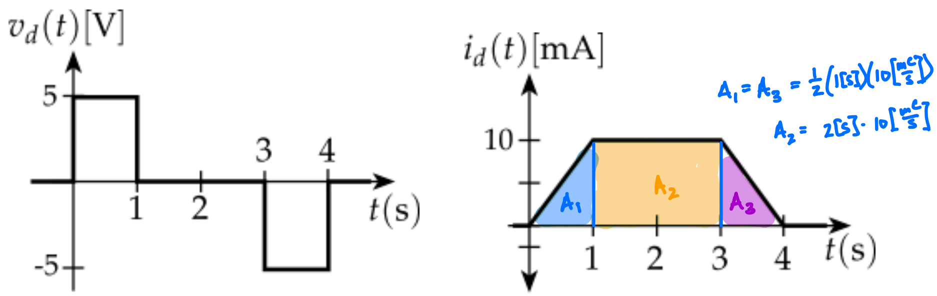


Figure B

(a)

$$i_d(t) = \begin{cases} 10[\frac{mA}{s}]t & 0<t<1[s] \\ 10[mA] & 1[s]<t<3[s] \\ -10[\frac{mA}{s}]t + 40[mA] & 3[s]<t<4[s] \\ 0 & t<0 \text{ or } t>4[s] \end{cases}$$

$$v_d(t) = \begin{cases} 5[V] & 0<t<1[s] \\ 0 & t<0 \text{ or } 1[s]<t<3[s] \text{ or } t>4[s] \\ -5[V] & 3[s]<t<4[s] \end{cases}$$

(b) $i = \frac{dq}{dt} \Rightarrow dq = i\,dt \Rightarrow Q = \int i\,dt$

$$Q = \int_0^{1[s]} 10[\tfrac{mA}{s}]t\,dt + \int_{1[s]}^{3[s]} 10[mA]\,dt + \int_{3[s]}^{4[s]} (-10[\tfrac{mA}{s}]t + 40[mA])\,dt$$

$$= 5[\tfrac{mA}{s}]t^2\Big|_0^{1[s]} + 10[mA]t\Big|_{1[s]}^{3[s]} + (-5[\tfrac{mA}{s}]t^2)\Big|_{3[s]}^{4[s]} + 40[mA]t\Big|_{3[s]}^{4[s]}$$

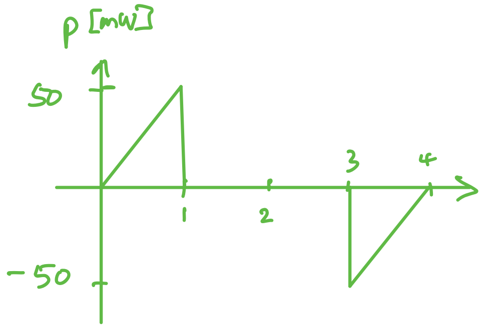
$$= 5[mC] + 30[mC] - 10[mC] - 80[mC] + 45[mC] + 160[mC] - 120[mC]$$

$$Q = 30[mC]$$

2 × triangles + 1 rectangle

or use area : $2(\frac{1}{2}b_1h) + b_2h = 1[s]\cdot 10[mA] + 2[s]\cdot 10[mA] = 30[mC]$

(c) Since $p = i \cdot v$



(d) $p = \frac{dw}{dt} \Rightarrow dw = p\,dt \Rightarrow w = \int p(t)\,dt$

Since area under each curve is the same, $W = 0$

Alternatively, evaluate integrals:

$$w = \int_0^{1[s]} 50\left[\tfrac{mW}{s}\right]t + \int_{3[s]}^{4[s]} \left(50\left[\tfrac{mW}{s}\right]t - 200[mW]\right)dt$$

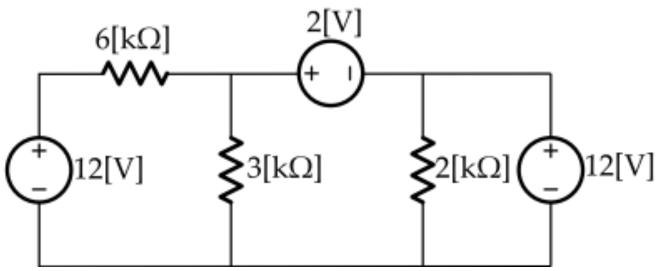
$$= 25\left[\tfrac{mW}{s}\right]t^2\Big|_0^{1[s]} + 25\left[\tfrac{mW}{s}\right]t^2\Big|_{3[s]}^{4[s]} - 200[mW]t\Big|_{3[s]}^{4[s]}$$

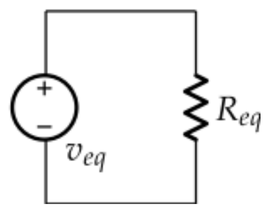
$$= 25[mJ] + 400[mJ] - 225[mJ] - 800[mJ] + 600[mJ]$$

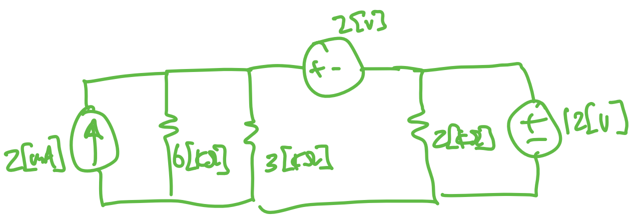
$$W = 0[mJ]$$

Problem 2

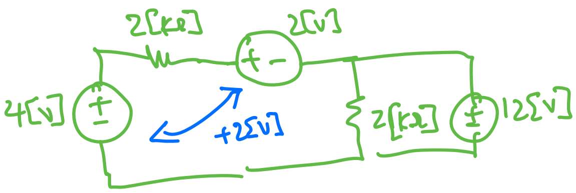
Use source transformations and series/parallel resistance combinations to simplify the circuit on the left into the equivalent circuit on the right to find v_{eq} and R_{eq}.

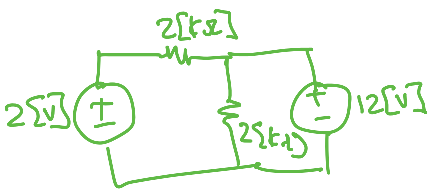


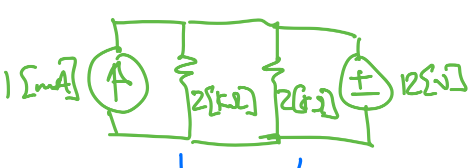




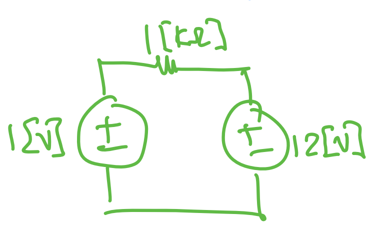
$\frac{3\cdot 6}{9}$[kΩ] = 2[kΩ]

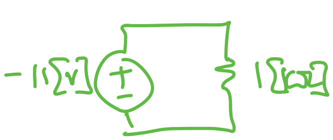






1[kΩ]





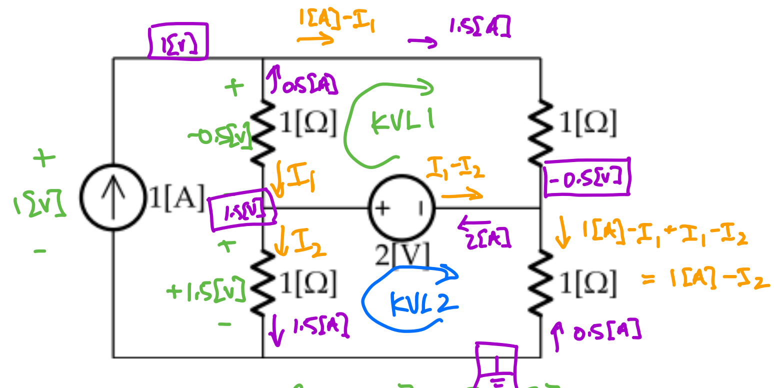
v_{eq} = -11[V]

R_{eq} = 1[kΩ]

Problem 3

For the circuit below

1. Write two raw KVL equations and any necessary KCL equations to solve this circuit
2. Simply the equations by collecting terms and write them into the space provided
3. Solve for the missing currents in each resistor
4. Determine the power supplied by the sources
5. Determine the power received by the resistors



KVL1 : $-I_1 \cdot 1[\Omega] + (1[A] - I_1)\cdot 1[\Omega] - 2[V] = 0$

$-2[\Omega] I_1 - 1[V] = 0$

$I_1 = -0.5[A]$

KVL2 : $-I_2 \cdot 1[\Omega] + 2[V] + (1[A] - I_2)1[\Omega] = 0$

$-2[\Omega] I_2 + 3[V] = 0$

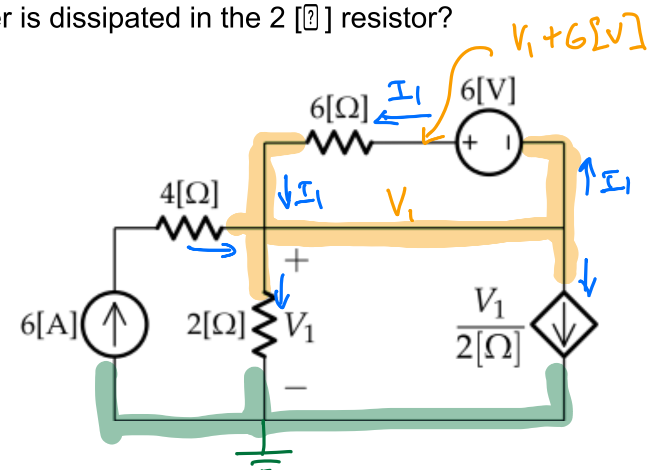
$I_2 = 1.5[A]$

Raw equations	$-I_1 \cdot 1[\Omega] + (1[A] - I_1)\cdot 1[\Omega] - 2[V] = 0$ $-I_2 \cdot 1[\Omega] + 2[V] + (1[A] - I_2)1[\Omega] = 0$
Simplified equations	$I_1 = -0.5[A]$ $I_2 = 1.5[A]$
Currents	$I_1 = -0.5[A]$ $I_2 = 1.5[A]$
p (sources)	$(1.5[V] + (-0.5[A]))\cdot 1[A] = 1[W]$ $\quad 1[W] + 4[W] = 5[W]$ $-(-0.5[A] - 1.5[A])2[V] = 4[W]$
p (resistors)	$((-0.5[A])^2 + (1.5[A])^2 + (1.5[A])^2 + (0.5[A])^2)\cdot 1[\Omega] =$ $0.25[W] + 2.25[W] + 2.25[W] + 0.25[W] = 5[W]$

Problem 4

Using nodal analysis (i.e., node voltage method),

a. Label nodes needed to solve this problem and write the two raw node voltage equation(s) in the space provided
b. Simplify the equation(s) and write them in the space provided
c. Determine the values of the node voltages
d. How much power is dissipated in the 2 [?] resistor?



$$6[A] + I_1 = \frac{V_1}{2[\Omega]} + \frac{V_1}{2[\Omega]} + I_1$$

$$2V_1 = 12[V] \Rightarrow V_1 = 6[V]$$

$$I_1 = \frac{V_1 + 6[V] - V_1}{6[\Omega]} = 1[A]$$

$$P = \frac{V_1^2}{2[\Omega]} = \frac{36}{2}[W] = 18[W]$$

Raw equation(s)	$6[A] + I_1 = \frac{V_1}{2[\Omega]} + \frac{V_1}{2[\Omega]} + I_1$	optional: $I_1 = \frac{V_1 + 6[V] - V_1}{6[\Omega]}$
Simplified equation(s)	$V_1 = 6[V]$	$I_1 = 1[A]$
Node Voltage(s)	$V_1 = 6[V]$	

p for 2 [?] resistor	18[W]